DocInsights Workshop at EMNLP 2026
Shared Task, Dr.DocBench Track

Consensus Selection over Prompt-Diverse VLM Transcriptions for Expert-Level Document Parsing

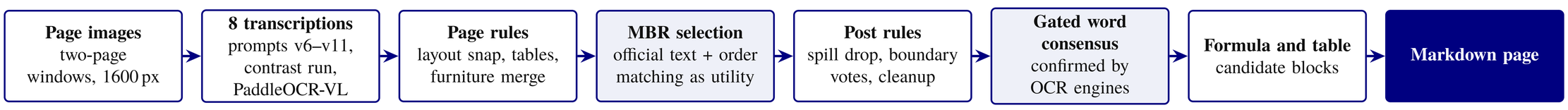
Pranshu Rastogi
Independent Researcher
rastogipranshu29@gmail.com

Rajneesh Tiwari
Independent Researcher
rajneesh.vish1@gmail.com

Scan for the full report
pranshurastogi29.github.io/drdocbench

7th	84.54	0.1641	85.65	81.84	71.86
final leaderboard	Overall	Text_ED	Reading order	Formula CDM	Table TEDS

509 test pages · training-free · one free vision-language model, prompted many ways · about $82 of API use in total



1 The Challenge

Dr.DocBench (Yang et al., 2026), a track of the DocInsights 2026 shared task (DocInsights Workshop Organizers, 2026), collects book pages on which parsers disagree. Its references follow **book-specific conventions** (which lines form a block, whether picture text is kept, where page numbers go), so a strong model loses points on pages it reads correctly.
Scoring uses the md2md matcher of OmniDocBench (Ouyang et al., 2025): text edit distance, reading order (RO), CDM (Wang et al., 2025) and TEDS (Zhong et al., 2020); Overall averages them per page.

Scorer facts that shaped the system

- Predicted blocks are joined to one reference block only if it has at most 300 normalised characters.
- Only word characters count: punctuation and Markdown emphasis do not; case and digits do.
- Formulas and tables are scored by best match, apart from text and order.

2 Select, Don't Train

One model gives many plausible transcriptions. We pick one per page with minimum Bayes risk (Kumar and Byrne, 2004), using the **metric itself as the utility**:

$$\hat{c} = \arg\max_{c \in C} \frac{1}{|R_c|} \sum_{r \in R_c} \left(w_t \, T(c \mid r) + w_o \, O(c \mid r) \right)$$

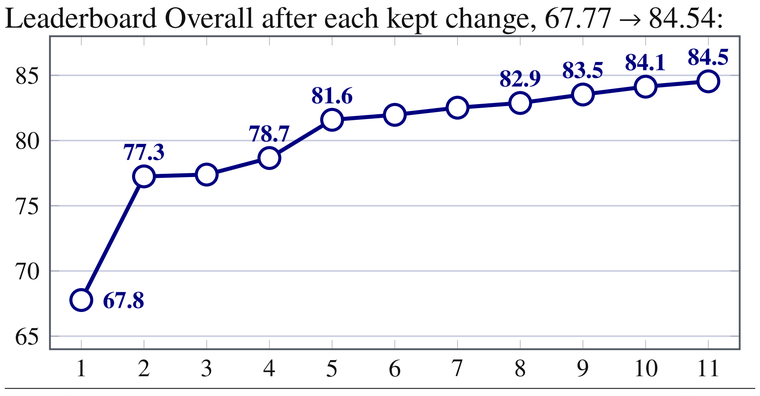
T and O are the official text and order scores of c with run r as the reference, R_c excludes c, and $(w_t, w_o) = (0.558, 0.433)$, the weights of text and order in Overall fitted on our submissions.
mbr8: eight candidates (five base runs incl. PaddleOCR-VL (Cui et al., 2025), three column-break runs) against eleven pseudo-references.

3 Gated Word Consensus

In the spirit of ROVER (Fiscus, 1997), the other runs vote word by word. A word changes only if a strict majority agrees, the new word is a near-spelling of the old one (not case only, not at a paragraph edge), and PaddleOCR-VL or MinerU (Wang et al., 2024) confirms it or the old word is not in a dictionary.
A plain majority vote lost **0.06**: our runs share one model's errors. The gated vote gains **+0.006** with no page worse and fixes 33 test words, e.g. *ATHAWAY* → *HATHAWAY*.

4 Development Journey

Leaderboard Overall after each kept change, 67.77 → 84.54:



#	Change kept	Overall
1	Single run, prompt v3	67.77
2	Furniture-last prompts, medoid of runs, layout snap	77.25
3–4	PaddleOCR-VL tables; contrast run as a vote	78.66
5–7	Formula and table candidate blocks	82.52
8	**MBR selection**, furniture merge, spill drop, boundary votes	82.88
9	More formula candidates, symbol channel	83.53
10	**Column-break runs in MBR**, word consensus	84.13
11	**Final**: more candidate sources	**84.54**

5 Ablations

Leave-one-out on 925 development pages (Δ in leaderboard units):

Pipeline	Text_ED	RO	Δ
Final (mbr8 + rules)	0.1168	88.58	–
– layout snap, tables	0.1204	87.11	**−0.84**
– furniture merge	0.1162	87.49	**−0.44**
– boundary votes	0.1186	88.48	**−0.14**
– spill drop	0.1187	88.55	**−0.12**
– cleanup	0.1169	88.58	**−0.01**
+ word consensus	0.1168	88.58	**+0.005**

Selection

- MBR beats the medoid by **+1.05** (book-clustered CI [+0.38, +1.85]).
- Learned rankers, weighted MBR and paragraph fusion: within noise of MBR.
- A per-page oracle over the same runs adds +4.8: selection, not reading, is the limit.
- Column-break runs as candidates: **+0.88** on dev, **+0.60** on the leaderboard.

6 What Transferred to Test

Selection did: mbr8 kept 68% of its development gain (94% for reading order). **Convention rules did not**:

Change (Overall)	Dev	Test
Contrast runs as extra votes	**+0.39**	**−0.84**
One-page windows as the pool	**+1.02**	**−0.63**
GPT-6.1 on 120 hard pages	**+0.06**	**−0.47**
Drop text PaddleOCR-VL lacks	**+0.02**	**−0.21**
Column splits incl. lists	**+1.00**	**−0.02**

7 Where the Error Is

Diagnostic submissions blanked groups of test books; the score change decodes each group:

Test books	Pages	Text_ED	RO
Subjects absent from dev	120	0.200	77.4
Highest run disagreement	120	0.203	80.7
Half of the rest	134	0.157	85.0
Other half	135	0.120	87.5

The first two groups hold **56%** of the text error. On dev, half of it sits on 8% of pages with a book-specific reference choice.

8 Stronger Models Alone

53 common dev pages	Text_ED	RO	Δ
Our pipeline (mbr8)	0.0857	88.45	–
Kimi K2.5	0.0614	85.20	**−0.05**
Gemini 3.8 Flash	0.0682	84.90	**−0.56**
Kimi K2.6, no thinking	0.1053	83.71	**−3.14**
GPT-6 sol-pro (897 pages)	0.1380	83.69	**−3.19**

Kimi K2.5 (Kimi Team et al., 2026) reads text best but orders worse; proofreading our pages with it: **−2.6**.

Takeaways

- Prompt diversity plus MBR with the metric as utility beat every stronger single model we tried.
- Selection transfers to the hidden test set; rules tuned to development books do not.
- The remaining gap is in segmentation and order conventions that vary by book.

Models and tools. Base transcriber: a free OpenRouter model (OpenRouter, 2026b). Also in the final submission: GPT-6.1 sol-pro (OpenAI, 2026c), as a source of formula and table candidates. Tested: GPT-6 sol-pro and luna-pro (OpenAI, 2026b,a), Gemini 3.8 Flash (Google, 2026), Claude Sonnet 5.5 (Anthropic, 2026), Kimi K2.6 (Moonshot AI, 2026), Qwen3.8 (Qwen Team, 2026), GLM-5.3 Flash (Z.ai, 2026), DeepSeek V4.1 Flash (DeepSeek, 2026), MiMo v2.6 Pro (Xiaomi, 2026) and dots-3-note-preview (dots studio, 2026), through OpenRouter (OpenRouter, 2026a) and Hugging Face Inference Providers (Hugging Face, 2026). Training-free: no fine-tuning, no external annotated data.

Links. Final ranking: https://drdocbench-challenge.abaka-pages.com/#leaderboard · Report: https://pranshurastogi29.github.io/drdocbench/ · Paper: https://pranshurastogi29.github.io/drdocbench/paper.pdf · Code (MIT): https://github.com/pranshurastogi29/drdocbench-challenge

References

Anthropic. 2026. Claude Sonnet 5.5. Model `anthropic/claude-sonnet-5.5`, accessed through OpenRouter, October 2026.

Cheng Cui, Ting Sun, Suyin Liang, Tingquan Gao, Zelun Zhang, Jiaxuan Liu, et al. 2025. PaddleOCR-VL: Boosting multilingual document parsing via a 0.9B ultra-compact vision-language model. *arXiv preprint arXiv:2510.14528*.

DeepSeek. 2026. DeepSeek V4.1 Flash. Model `deepseek/deepseek-v4.1-flash`, accessed through OpenRouter, October 2026.

DocInsights Workshop Organizers. 2026. DocInsights 2026 shared task: DocSem and Dr.DocBench tracks. Workshop on Document Intelligence and Understanding at EMNLP 2026. https://docinsights-workshop.github.io/docinsights-2026/shared-task/.

dots studio. 2026. dots-3-note-preview. Model `dots-studio/dots-3-note-preview`, accessed through OpenRouter, October 2026.

Jonathan G. Fiscus. 1997. A post-processing system to yield reduced word error rates: Recognizer output voting error reduction (ROVER). In *Proceedings of the IEEE Workshop on Automatic Speech Recognition and Understanding*, pages 347–354.

Google. 2026. Gemini 3.8 Flash. Model `google/gemini-3.8-flash`, accessed through OpenRouter, October 2026.

Hugging Face. 2026. Inference providers. https://huggingface.co/docs/inference-providers.

Kimi Team, Tongtong Bai, Yifan Bai, Yiping Bao, S. H. Cai, Yuan Cao, et al. 2026. Kimi K2.5: Visual agentic intelligence. *arXiv preprint*.

Shankar Kumar and William Byrne. 2004. Minimum Bayes-risk decoding for statistical machine translation. In *Proceedings of the Human Language Technology Conference of the North American Chapter of the Association for Computational Linguistics: HLT-NAACL 2004*, pages 169–176.

Moonshot AI. 2026. Kimi K2.6. Model `moonshotai/Kimi-K2.6`, accessed through Hugging Face Inference Providers, October 2026.

OpenAI. 2026a. GPT-6 luna-pro. Model `openai/gpt-6-luna-pro`, accessed through OpenRouter, September 2026.

OpenAI. 2026b. GPT-6 sol-pro. Model `openai/gpt-6-sol-pro`, accessed through OpenRouter, October 2026.

OpenAI. 2026c. GPT-6.1 sol-pro. Model `openai/gpt-6.1-sol-pro`, accessed through OpenRouter, October 2026.

OpenRouter. 2026a. OpenRouter. https://openrouter.ai.

OpenRouter. 2026b. space-bunny-alpha (stealth model). Model `stealth/space-bunny-alpha`, offered free on OpenRouter until October 2026; developer not disclosed.

Linke Ouyang, Yuan Qu, Hongbin Zhou, Jiawei Zhu, Rui Zhang, Qunshu Lin, et al. 2025. OmniDocBench: Benchmarking diverse PDF document parsing with comprehensive annotations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Qwen Team. 2026. Qwen3.8 models. Models `qwen/qwen3.8-flash` and `qwen/qwen3.8-27b`, accessed through OpenRouter, September 2026.

Bin Wang, Fan Wu, Linke Ouyang, Zhuangcheng Gu, Rui Zhang, Renqiu Xia, Botian Shi, Bo Zhang, and Conghui He. 2025. Image over text: Transforming formula recognition evaluation with character detection matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19681–19690.

Bin Wang, Chao Xu, Xiaomeng Zhao, Linke Ouyang, Fan Wu, Zhiyuan Zhao, et al. 2024. MinerU: An open-source solution for precise document content extraction. *arXiv preprint arXiv:2409.18839*.

Xiaomi. 2026. MiMo v2.6 Pro. Model `xiaomi/mimo-v2.6-pro`, accessed through OpenRouter, October 2026.

Minglai Yang, Xinyan Velocity Yu, Pengyuan Li, Xinyu Guo, Zhenting Qi, Konwoo Kim, et al. 2026. Dr.DocBench: A comprehensive benchmark for expert-level and difficult document parsing. *arXiv preprint arXiv:2606.01393*.

Z.ai. 2026. GLM-5.3 Flash. Model `z-ai/glm-5.3-flash`, accessed through OpenRouter, October 2026.

Xu Zhong, Elaheh ShafieiBavani, and Antonio Jimeno Yepes. 2020. Image-based table recognition: Data, model, and evaluation. In *European Conference on Computer Vision (ECCV)*, pages 564–580.